SAT-Style Practice Test

Original Composition — Full-Length Adaptive Edition

Section 1: Reading and Writing (Module 1, Module 2 Easier, Module 2 Harder)

Section 2: Math (Module 1, Module 2 Easier, Module 2 Harder)

Total Questions: 147

Adaptive Structure

Section	Module 1 (All Students)	Module 2 Easier	Module 2 Harder
Reading & Writing	27 Qs	27 Qs (~70% E/M, ~30% H)	27 Qs (~30% E/M, ~70% H)
Math	22 Qs	22 Qs (~70% E/M, ~30% H)	22 Qs (~30% E/M, ~70% H)

Answer key not included — to be provided separately upon request.

SECTION 1: READING AND WRITING

Module 1 (27 Questions — All Students)

Question 1

Skill: Vocabulary in Context | Difficulty: Easy

The following text is from Zora Neale Hurston's 1937 novel *Their Eyes Were Watching God*. The narrator describes the protagonist Janie's experience during a severe hurricane in the Florida Everglades: "The wind came back with triple fury, and put out the light for the last time. They sat in company with the others in other shanties, their eyes straining against crude walls and their souls asking if He meant to measure their puny might against His. They seemed to be staring at the dark, but their eyes were watching God." The passage conveys a mood of profound **reverence** even amid destruction.

- A) Worship
- B) Awe
- C) Obedience
- D) Gratitude

Question 2

Skill: Vocabulary in Context | Difficulty: Easy

The following text is adapted from a 1901 translation of Leo Tolstoy's 1877 novel *Anna Karenina* (translated by Constance Garnett). In the text, Levin is walking through a birch forest in early spring. The morning air was saturated with moisture, and every leaf and blade of grass seemed to hold the night's rain in trembling droplets. The silence was so complete that Levin felt the forest was holding its breath, waiting for the first ray of sun to **dispel** the grey veil of dawn.

- A) Ignore
- B) Scatter
- C) Confirm
- D) Observe

Question 3

Skill: Vocabulary in Context | Difficulty: Medium

When marine biologists discover a previously unidentified deep-sea organism, it is first assigned a temporary code based on the expedition number and sample location. After the specimen's morphology has been confirmed through laboratory analysis, the lead researcher may propose a formal binomial name. Under current International Commission on Zoological Nomenclature guidelines, it is ______ for researchers to name organisms after mentors or colleagues, but they are not required to do so; the final classification decision rests with the relevant review panel.

- A) customary
- B) impractical
- C) controversial
- D) obligatory

Question 4

Skill: Vocabulary in Context | Difficulty: Medium

The elaborate jade burial masks recovered from ancient Maya tombs at Palenque, Mexico, demonstrate the civilization's mastery of lapidary arts, a tradition that is ______ across Mesoamerica—archaeologists excavating Copán in Honduras, for instance, can readily observe how its jade ornaments draw from the same carving conventions.

A) documented
B) pervasive
C) challenged
D) exaggerated

Question 5

Skill: Text Structure and Purpose | Difficulty: Medium

The following text is from Nikolai Gogol's 1842 novel *Dead Souls* (translated by Richard Pevear and Larissa Volokhonsky in 1996). The narrator is describing a provincial governor's ball.

A stout gentleman in a military uniform appeared at precisely nine o'clock and positioned himself at the center of the refreshment table. He surveyed the assembled guests with the satisfied expression of a general inspecting newly polished cannons, as if to announce, "I have attended balls in Moscow and Petersburg, and you should count yourselves fortunate that I have deigned to appear at this insignificant gathering." There is a certain breed of official who treats every provincial event as a stage for demonstrating metropolitan superiority.

A) It reveals the stout gentleman's desire to become the governor of the province.
B) It shows the narrator's sympathy for the provincial guests who feel intimidated by the gentleman.
C) It illustrates the gentleman's patronizing attitude toward the other attendees at the ball.
D) It explains why certain officials avoid provincial gatherings entirely.

Question 6

Skill: Text Structure and Purpose | Difficulty: Medium

One Hundred Years of Solitude, first published in 1967, is a novel in Gabriel García Márquez's body of work, which encompasses numerous novels and short story collections. Like much of Márquez's fiction, *One Hundred Years of Solitude* features a densely layered timeline, but the companion novella *Chronicle of a Death Foretold* revisits some of the same narrative techniques. Thus, the complexity of the timeline is only an issue if one reads the works in isolation—<u>Márquez's body of work is best understood as an interconnected literary universe, similar to William Faulkner's Yoknapatawpha novels</u>.

A) It demonstrates that readers who begin with *Chronicle of a Death Foretold* will prefer it to *One Hundred Years of Solitude*.
B) It argues that Márquez's fiction deserves comparison with Faulkner's on the basis of critical acclaim.
C) It offers a perspective that helps explain why a certain characteristic of *One Hundred Years of Solitude* need not be seen as a weakness.
D) It presents evidence that *One Hundred Years of Solitude* and *Chronicle of a Death Foretold* are Márquez's most commercially successful books.

Question 7

Skill: Text Structure and Purpose | Difficulty: Hard

Brazilian landscape architect Roberto Burle Marx's influential career, which spanned the 1930s to the 1990s, evolved through distinct phases. As exemplified by the Copacabana Beach promenade, many of Burle Marx's early projects introduced bold, abstract mosaic patterns into public spaces. <u>Extended botanical expeditions into the Amazon basin later sparked an engagement with native plant ecosystems</u>—designs whose emphasis on indigenous flora, ecological balance, and sculptural planting beds and whose departure from purely ornamental commissions are readily apparent in Burle Marx's Ibirapuera Park gardens in São Paulo.

A) It presents biographical information that accounts for the shift in Burle Marx's career that is discussed in the text.
B) It notes why the two landscape projects described in the text constitute major departures from Burle Marx's typical style.
C) It lends support to an argument about the abstract mosaic aesthetic that is made in the previous sentence.
D) It explains why the landscape project discussed later in the sentence was widely celebrated by Burle Marx's peers.

Question 8

Skill: Central Ideas and Details | Difficulty: Medium

Senegalese filmmaker Ousmane Sembène transforms cinematic narratives using his distinctive method of weaving oral storytelling traditions into scripted dialogue between actors from local communities. In some films, Sembène focuses on critiquing colonial legacies, as he does in his portrayal of a labor strike in Dakar. However, in other films, Sembène celebrates everyday village life, as he does in his vivid depiction of a grandmother navigating a bustling market. Sembène views both of these approaches as ways of representing the dignity and collective spirit of West African communities.

A) Sembène began his filmmaking career by critiquing colonialism and then shifted exclusively to depicting everyday village scenes.
B) Sembène's portrayal of a grandmother in a market emphasizes communal bonds more than individual hardship.
C) Sembène's films honor both significant historical struggles and ordinary moments of communal life as expressions of collective dignity.
D) Sembène's depiction of a labor strike is considered more politically significant than his depiction of the grandmother in the market.

Question 9

Skill: Central Ideas and Details | Difficulty: Medium

Why do some species of bats rely on echolocation while others depend primarily on vision? Researchers hypothesize that this divergence may be partly attributable to the structure of the larynx, a specialized organ in the throat. Insectivorous bats, which typically hunt in darkness, have a highly developed larynx that vibrates rapidly when air is expelled, producing ultrasonic pulses. By contrast, fruit bats have a relatively simple larynx and instead rely on large eyes and keen night vision. These laryngeal traits allow insectivorous bats and many other microbat species to navigate dense forests at night. The same traits may also reduce the evolutionary pressure on these bats to develop acute vision.

A) Both insectivorous bats and fruit bats have larynxes, but insectivorous bats' are complex whereas fruit bats' are simple.
B) Insectivorous bats and most other microbat species can echolocate because they have highly developed larynxes.
C) Researchers are uncertain why relatively small bats, such as insectivorous bats, echolocate but relatively large bats, such as fruit bats, use vision.
D) Differences related to the larynx in insectivorous bats and fruit bats may help explain why one species echolocates and the other relies on vision.

Question 10

Skill: Central Ideas and Details | Difficulty: Hard

In their systematic review of research on gamified fitness applications (apps that use game-like elements such as points, leaderboards, and challenges to encourage physical activity), Dr. Tomás Reyes et al. confirm that such apps, though they can initially boost user motivation, may not translate into sustained long-term behavioral change. This phenomenon can be explained by the overjustification effect as it is described in Edward Deci's self-determination theory. In this case, users' intrinsic motivation to exercise is gradually undermined by the external rewards the app provides, leaving little internal drive to continue once the novelty of the game elements wears off.

A) Deci's self-determination theory provides a framework for explaining the finding by Reyes et al. that gamified apps may ultimately diminish users' intrinsic motivation to exercise.
B) The systematic review by Reyes et al. reveals that higher engagement in gamified app features is linked to lower long-term adherence to fitness routines.
C) Although Deci's self-determination theory suggests otherwise, gamified fitness apps can succeed as motivational tools, provided they balance game elements and intrinsic rewards.
D) Research by Reyes et al. supports Deci's conclusion that because people primarily focus on external rewards, it is impossible for gamified apps to sustain motivation over time.

Question 11

Skill: Command of Quantitative Evidence | Difficulty: Easy

A student is writing a research paper on the adoption of electric vehicle models in Europe during the 2010s and 2020s. The student wants to determine the approximate number of units sold in Europe of the Renault Zoe, manufactured by Renault. The student finds from the table that the number sold was approximately ______

[PLACEHOLDER: Table titled 'Electric Vehicle Sales in Europe, 2012–2022.' Columns: Vehicle, Manufacturer, Vehicle Type, Approx. Units Sold in Europe. Rows: Nissan Leaf / Nissan / battery electric / 310,000; Renault Zoe / Renault / battery electric / 485,000; Tesla Model 3 / Tesla / battery electric / 1,120,000; BMW i3 / BMW / battery electric / 78,000.]

A) 485,000 units.
B) 310,000 units.
C) 1,120,000 units.
D) 78,000 units.

Question 12

Skill: Command of Quantitative Evidence | Difficulty: Medium

A student is researching global wheat production. The student learns that the total amount of wheat harvested worldwide increased from 585 million metric tons in 2005 to 779 million metric tons in 2021. Considering a table with information about four countries, the student notes that the global increase in wheat production occurred even though ______

[PLACEHOLDER: Table titled 'Millions of Metric Tons of Wheat Harvested in 2005 and 2021.' Columns: Country, 2005, 2021. Rows: Australia / 25.4 / 21.7; India / 68.6 / 109.6; France / 36.9 / 36.1; Argentina / 16.0 / 22.1.]

A) each of the four countries produced less than 50 million metric tons in 2005.
B) Australia harvested less wheat in 2021 than in 2005.
C) India harvested more wheat in 2021 than in 2005.
D) each of the four countries produced more than 25 million metric tons in 2021.

Question 13

Skill: Textual Evidence | Difficulty: Medium

Middlemarch is an 1871 novel by George Eliot set in a fictional English Midlands town. In the novel, the idealistic young scholar Dorothea Brooke agrees to a marriage with the much older clergyman Mr. Casaubon, which the narrator attributes to Dorothea's earnest but misguided intellectual ambition: ______

A) "The really delightful marriage must be that where your husband was a sort of father, and could teach you even Hebrew, if you wished it."
B) "In every province of human thought there are moments when the eye of wisdom rests upon the horizon of knowledge."
C) "Dorothea's fervent desire for learning was so consuming that she mistook the dryness of Casaubon's scholarship for the waters of profound thought, and she accepted his proposal as one might accept admission to a great library."
D) "It is a narrow mind which cannot look at a subject from various points of view."

Question 14

Skill: Command of Evidence (Strengthen/Weaken) | Difficulty: Hard

Singapore has deployed a network of autonomous cleaning robots across 60% of its public housing estates to reduce maintenance costs and improve sanitation standards. Investigating the impact of such robotic systems on municipal cleaning efficiency in mid-sized South American cities, Dr. Rafael Mendoza et al. hypothesized that cleaning robots reduce labor hours more effectively than they reduce overall waste accumulation. To evaluate this, the researchers tracked labor deployment data and waste tonnage across ten pilot neighborhoods, using a Sanitation Efficiency Index (SEI), on which a high score corresponds to high efficiency.

A) Pilot neighborhoods with a high percentage of robots and low manual staffing had higher average SEI scores than did pilot neighborhoods with fewer robots but similar total cleaning capacity.
B) Pilot neighborhoods in tropical climates with high robot density had lower average SEI scores than did pilot neighborhoods in similar climates with lower robot density.
C) Pilot neighborhoods with both high robot coverage and high manual staffing had similar SEI scores, whereas pilot neighborhoods with low coverage of both had widely differing SEI scores.
D) Pilot neighborhoods where cleaning robots were deployed showed significantly reduced labor hours but only marginal decreases in total waste tonnage, compared to control neighborhoods without robots.

Question 15

Skill: Inferences | Difficulty: Hard

All commercial fertilizers contain varying combinations of nitrogen, phosphorus, and potassium. However, synthetic fertilizers, commonly used in large-scale agriculture, contain a higher proportion of nitrogen (at least 20%) than do organic compost blends, which is responsible for their faster nutrient uptake. Unlike synthetic fertilizers, organic compost blends have a granular structure that relies on a different decomposition chemistry. Organic compost blends have two main variants: the standard type, often used in home gardens, and the enriched type, which releases nutrients more slowly and is commonly used in commercial orchards. Thus, fertilizers used in ______

A) large-scale agriculture will have faster nutrient uptake than fertilizers used in commercial orchards will.
B) home gardens will have a nitrogen proportion lower than 20%, while fertilizers used for large-scale agriculture will not.
C) large-scale agriculture will have a granular decomposition structure.
D) commercial orchards will likely have a similar decomposition chemistry to fertilizers used in home gardens.

Question 16

Skill: Standard English Conventions | Difficulty: Easy

In Portland, Oregon, protected pedestrian bridges like the Tilikum Crossing—which spans approximately 1,720 feet over the Willamette River—have become an increasingly popular means of transit for both commuters and tourists alike. Moreover, Marcus ______ has identified expanded pedestrian infrastructure as one key strategy to reduce vehicle emissions, noting the city's mild autumns and extensive public transit network.

A) Okonkwo an urban planner and sustainability advocate
B) Okonkwo, an urban planner and sustainability advocate,
C) Okonkwo an urban planner and sustainability advocate,
D) Okonkwo, an urban planner and sustainability advocate

Question 17

Skill: Standard English Conventions | Difficulty: Medium

The Great Wall in northern China and the Roman Hadrian's Wall in northern England are two of several ancient ______ massive defensive barriers. A distinct pattern of erosion affects the ramparts of the Great Wall, while a different pattern of erosion affects the stones of Hadrian's Wall.

A) sites, that feature their own historic fortifications, and a
B) sites that feature their own historic fortifications. A
C) sites, that feature their own historic fortifications and a
D) sites that feature their own historic fortifications, a

Question 18

Skill: Standard English Conventions | Difficulty: Medium

With one observatory in Chile and another in Hawaii, the astronomical research stations that form the Global Telescope Array—or GTA, as astrophysicists like Dr. Ingrid Larsson call it—number nearly seventy ______ across the globe, the telescopes capture light from distant galaxies and other celestial phenomena.

A) stations. Positioned
B) stations, positioned
C) stations, which are positioned
D) stations positioned

Question 19

Skill: Standard English Conventions | Difficulty: Medium

"Paradox," like other three-syllable words that pair exclusively with one other word (e.g., "orthodox" and "unorthodox"), ______ with "equinox" to form a closed rhyming pair.

A) combine
B) are combined
C) combines
D) have combined

Question 20

Skill: Standard English Conventions | Difficulty: Hard

As varied as their disciplines may seem, connections exist between Marie Curie, Polish-French physicist of the Sorbonne ______ Flemish anatomist of the Renaissance tradition; and Hypatia, Greek mathematician of the Alexandrian era.

A) School, Andreas Vesalius,
B) School; Andreas Vesalius,
C) School, Andreas Vesalius;
D) School Andreas Vesalius,

Question 21

Skill: Standard English Conventions | Difficulty: Hard

Photosynthesis, ______ widely understood to convert solar energy into chemical energy in a variety of plant organisms—including algae, mosses, and flowering plants—provides similar metabolic functions, albeit through different biochemical pathways, to certain chemosynthetic bacteria found near deep-sea hydrothermal vents.

A) photosynthesis is
B) photosynthesis, is
C) photosynthesis has been
D) photosynthesis.

Question 22

Skill: Transitions | Difficulty: Easy

In 1938, Orson Welles's radio broadcast of *The War of the Worlds* became one of the first dramatic performances to provoke widespread public reaction in the United States. ______ many other works of broadcast fiction generated significant audience responses, including Rod Serling's *The Twilight Zone* in 1959 and the BBC's *Ghostwatch* in 1992.

A) That is,
B) Previously,
C) For example,
D) Later,

Question 23

Skill: Transitions | Difficulty: Medium

The element helium is remarkably stable, consistently ranking as one of the least reactive elements on the periodic table (helium is listed as entirely inert in most standard references). By contrast, the reactivity of the alkali metal potassium varies considerably. Indeed, when exposed to humid air between June and September 2022, potassium samples oxidized with a striking increase in speed; ______ their tarnishing rate accelerated from gradual to nearly instantaneous.

A) in sum,
B) in turn,
C) specifically,
D) for example,

Question 24

Skill: Transitions | Difficulty: Easy

Generally, ocean-dwelling mammals dive to greater depths than freshwater mammals. The sperm whale, ______ can descend to depths exceeding 7,000 feet during hunting dives, while a typical river otter rarely submerges below 60 feet.

A) however,
B) meanwhile,
C) for example,
D) additionally,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• The Pont du Gard in southern France is a former Roman aqueduct, constructed across a river gorge to carry water to the city of Nîmes.

• The Aqueduct of Segovia in central Spain is another former Roman water-delivery structure.

• It was constructed on a granite ridge above the city and remained in continuous use until approximately 1973.

• Roman aqueduct engineering influenced the development of modern hydraulic infrastructure.

The student wants to contrast the Pont du Gard and the Aqueduct of Segovia. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Pont du Gard was built across a river gorge, whereas the Aqueduct of Segovia was built on a granite ridge above a city.
B) In contrast to the Pont du Gard, the Aqueduct of Segovia was built across a river gorge to carry water.
C) Although they were both built by the Romans, the Pont du Gard and the Aqueduct of Segovia are no longer in active service.
D) Unlike the Pont du Gard, the Aqueduct of Segovia was constructed by Roman engineers, the predecessors of modern hydraulic engineers.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

• The Pritzker Architecture Prize is an international award administered by the Hyatt Foundation in Chicago, Illinois.

• Architect Zaha Hadid of Iraq and the UK won a Pritzker Prize in 2004.

• Hadid's recognized body of work included the Heydar Aliyev Center.

• Architect Diébédo Francis Kéré of Burkina Faso won a Pritzker Prize in 2022.

• Kéré's recognized body of work included the Serpentine Pavilion.

The student wants to emphasize a similarity between the two architects. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both Hadid and Kéré are Pritzker Architecture Prize-winning architects.
B) Hadid, an architect, won a Pritzker Prize in 2004 for works including the Heydar Aliyev Center.
C) The Hyatt Foundation in Chicago has awarded Pritzker Prizes to architects from both Iraq and Burkina Faso.
D) In contrast to Hadid's Heydar Aliyev Center, Kéré is known for the Serpentine Pavilion.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• Vineyards and other agricultural operations use irrigation systems to maintain optimal soil moisture levels.

• A common irrigation method is drip irrigation, which delivers water directly to the plant's root zone through a network of tubes.

• When a drip emitter releases water, it undergoes capillary action, distributing moisture laterally through the soil matrix.

• When the system is inactive, residual moisture dissipates through evaporation and root absorption.

• The irrigation method flood irrigation is a gravity-fed technique.

• Gravity-fed techniques rely on the natural slope of terrain to distribute water.

The student wants to describe the role of irrigation systems in vineyards. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Vineyards can distribute moisture and maintain soil levels through capillary action and evaporation.
B) The irrigation method flood irrigation, used in vineyards, relies on the natural slope of terrain.
C) Used in vineyards, drip emitters and other irrigation methods can distribute water and reclaim it through evaporation.
D) Vineyards use irrigation systems like drip irrigation to maintain optimal soil moisture by delivering water directly to root zones.

SECTION 1: READING AND WRITING

Module 2 — Easier Path (27 Questions)

Question 1

Skill: Vocabulary in Context | Difficulty: Easy

Although South American poetry is highly varied in its themes and structures, it is also characterized by notable thematic ______: scholars have traced parallels linking works by writers such as Pablo Neruda and César Vallejo to the indigenous oral traditions of the Quechua and Mapuche peoples.

A) fragility
B) refinement
C) continuity
D) novelty

Question 2

Skill: Vocabulary in Context | Difficulty: Easy

When discussing sculptural movements in Italy, art historians regularly characterize the works of Gian Lorenzo Bernini as especially representative of the Baroque era. The reasons for this characterization may seem _______, but associating Bernini exclusively with the Baroque risks overlooking the restraint in some of his later commissions that do not align with the exuberance of that artistic period.

A) incompatible
B) arrogant
C) inflammatory
D) undeniable

Question 3

Skill: Vocabulary in Context | Difficulty: Easy

The following text is adapted from a 1925 translation of Thomas Mann's 1924 novel *The Magic Mountain* (translated by H. T. Lowe-Porter). In the text, Hans Castorp has stepped onto the balcony of a sanatorium overlooking the Swiss Alps. And having reached the wooden railing he had to acknowledge that the afternoon light was softening, folding itself into longer shadows. Yes, the final brightness was thinning, stretching across the peaks, and there below, the valley, already dim, was apparent in pale wisps of mist drifting and curling, ever more delicate and transient, and destined no sooner formed than to be absorbed by the gathering dusk.

A) Positioned
B) Noticeable
C) Accomplished
D) Fading

Question 4

Skill: Vocabulary in Context | Difficulty: Medium

The soaring minarets and geometric tilework of the Blue Mosque in Istanbul, Turkey, exemplify the Ottoman contribution to Islamic architecture, an influence that is ______ throughout the Eastern Mediterranean—visitors touring the Great Mosque of Córdoba in Spain, for instance, can readily identify how its horseshoe arches draw inspiration from similar building traditions.

A) substantiated
B) evident
C) contested
D) minimized

Question 5

Skill: Text Structure and Purpose | Difficulty: Medium

In 2019, Dr. Yuki Tanaka and colleagues published a study concluding that rising ocean acidity has a measurable effect on the shell formation of Crassostrea gigas, a species of Pacific oyster. However, Tanaka and colleagues' study relied on a sample of only 12 specimens. In a 2022 review of various scientists' conclusions about the impacts of ocean acidification on bivalve development, Dr. Leila Hassan and colleagues caution that relying on such a small sample size can increase the potential for skewed results. <u>Such results, in turn, can contribute to reports of overstated effects.</u>

A) It summarizes a shift in scientists' understanding of how Crassostrea gigas has responded to ocean acidity.
B) It elaborates on a potential consequence of Tanaka and colleagues' reliance on a relatively small sample size.
C) It emphasizes the magnitude of the effect reported by Tanaka and colleagues of ocean acidity on Crassostrea gigas.
D) It counters the objection of Hassan and colleagues to studies that rely on relatively small sample sizes.

Question 6

Skill: Text Structure and Purpose | Difficulty: Medium

The following text is adapted from a 1908 translation of Guy de Maupassant's 1883 short story "A Piece of String." The narrator is describing Maître Hauchecorne, a peasant farmer in Normandy.

For the last fifteen years he had walked the same muddy path to the village square each Tuesday, and had nodded to the same merchants at the same hour, and nearly at the same stall. Every market day, after <u>arguing</u> over the price of harness leather, he <u>trudged</u> back up the hill like an old horse returning to its stable after a day of plowing, and <u>collapsed</u> onto the bench outside his door, always feeling diminished, as though the world were slowly forgetting him.

Taken together, the three underlined portions most clearly serve which function in the text as a whole?

A) They illustrate an unexplained anxiety that Hauchecorne experiences at the start of each market day.
B) They suggest Hauchecorne's exhaustion from being trapped in a life that has become monotonously repetitive.
C) They reveal that Hauchecorne is rigidly precise in his routine because of the expectations of his neighbors.
D) They convey Hauchecorne's growing nervousness about having failed to complete an important transaction.

Question 7

Skill: Central Ideas and Details | Difficulty: Easy

Rafflesia arnoldii is a species of parasitic plant found exclusively in the tropical rainforests of Sumatra and Borneo. Members of this species exhibit a remarkable range of adaptations, with some individuals producing flowers exceeding three feet in diameter to attract carrion flies, while others have reduced their vegetative body to thread-like filaments embedded in host vines. All specimens of Rafflesia descended from a single ancestral lineage that arrived in the region around 46 million years ago. The lineage's descendants diversified into distinct survival strategies as they adapted to the dense understory of the Southeast Asian rainforest.

A) To advance the claim that all parasitic plants in Southeast Asia belong to the Rafflesia genus and list their common ancestors

B) To identify the specific ancestral lineage of all parasitic plants and explain why they have such varied physical forms

C) To describe the Rafflesia species and explain how the plant family developed such varied survival strategies

D) To describe the specific habitat where Rafflesia arnoldii is found and identify other plants that share a common ancestor with it

Question 8

Skill: Cross-Text Connections | Difficulty: Medium

Text 1

One challenge faced by researchers studying global water quality is that countries may define water contamination differently. Many countries measure contamination based on parts per million of dissolved solids. However, countries do not all use the same thresholds; for example, Japan uses a maximum of 500 ppm, while Brazil uses 1,000 ppm. Other countries use a combination of dissolved solids and microbial counts. This variation makes it difficult for researchers to compare water quality data across different countries.

Text 2

Recently, a coalition of seven international environmental agencies developed standardized global classifications for levels of water contamination. The coalition developed a new framework called the "Unified Water Quality Index." This new framework establishes global criteria used to define five grades of water contamination (pristine, acceptable, marginal, polluted, and hazardous) and allows researchers to better compare global water quality conditions.

Based on the texts, how would the author of Text 2 most likely respond to the problem presented in Text 1?

A) By suggesting that researchers focus on topics besides water quality

B) By noting that a possible solution to the problem is available

C) By recommending that a specific institution should further investigate the problem

D) By supplying additional ways in which water quality research is difficult

Question 9

Skill: Inferences / Textual Evidence | Difficulty: Medium

Dr. Fatima al-Rashid and Dr. Wei Chen's research on the taxation policies of Empress Wu Zetian (who ruled the Tang Dynasty from 690 to 705) builds on other studies of East Asian early medieval state finance, including Twitchett's and Graff's work on the relationship between land taxation and imperial stability. But al-Rashid and Chen's unique contribution is their reconstruction of the earliest surviving complete set of provincial tax ledgers for any Tang ruler, which demonstrate that Wu Zetian's periodic revenue declines were caused by disruptions in provincial collection cycles, not systemic bureaucratic fraud.

A) Al-Rashid and Chen's research uses newly compiled evidence to rule out one possible explanation for Wu Zetian's revenue declines and to confirm another.
B) Al-Rashid and Chen's research revealed the role of Wu Zetian's revenue declines in exacerbating provincial unrest.
C) Al-Rashid and Chen's research builds on earlier work about Tang state finance by Twitchett and Graff and corrects errors in that earlier work.
D) Al-Rashid and Chen's research presented a novel body of evidence supporting Twitchett and Graff's hypothesis regarding why Wu Zetian faced recurring fiscal problems.

Question 10

Skill: Command of Evidence (Strengthen/Weaken) | Difficulty: Medium

Dr. Elena Petrova and Dr. Kwame Asante investigated the role of coloration in predator avoidance among butterfly species. They found that during controlled experiments, 38% of the tested species displayed conspicuous warning patterns, which ranged from bright orange bands in Danaus plexippus to vivid blue spots in Morpho menelaus. A second research team has claimed that butterflies use these patterns primarily to deter specific predator species.

A) Among the butterfly species that displayed warning patterns in response to predator presence, no individuals displayed patterns before predators arrived, but the vast majority displayed them once predators appeared.
B) Most of the butterfly species that were found not to display warning patterns in response to predator presence have been observed displaying similar patterns during mating rituals.
C) Butterflies that were found to display warning patterns in predator experiments are typically unpalatable and were tested in isolation.
D) The patterns butterflies displayed in response to predators had wavelength characteristics that would make them invisible to birds, lizards, or spiders—the most frequent predators of butterflies.

Question 11

Skill: Command of Evidence (Research Design) | Difficulty: Medium

Starbucks's expansion into bottled ready-to-drink beverages in 1996 is a well-known example of brand extension—the company leveraged its reputation as a coffeehouse chain to enter a product category where it had not previously competed. An outstanding question is whether perceived brand authenticity predicts consumers' likelihood of purchasing brand extensions. To answer this question, Dr. Sunita Kapoor et al. identified 30 extended-brand pairs (e.g., the same brand of coffee shops and bottled drinks) in 52 weeks of purchases by approximately 60,000 households and, for each pair, calculated the change in probability of a brand in one category being purchased if the same brand was purchased in the other category.

A) Have a representative sample of the households rate the authenticity of the brand in each extended-brand pair, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.
B) Poll a representative sample of the households to determine the degree of brand familiarity for each brand in the extended-brand pairs, then determine how, if at all, the degree of brand familiarity correlates with the average cost of each product in the pairs.
C) Have a representative sample of the households rate the quality of one product in each extended-brand pair relative to other products in the same category, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.
D) Poll a representative sample of the households to determine the degree of brand familiarity for each brand in the extended-brand pairs, then determine how, if at all, the degree of brand familiarity correlates with the frequency with which a different group of households purchased at least one product of that brand.

Question 12

Skill: Command of Quantitative Evidence | Difficulty: Medium

Education researcher Dr. Aisha Mbeki and colleagues developed a computational model to predict how engaged a student will be with a particular lecture on a scale from 1 (not at all) to 5 (very much). They then recruited participants to use the same scale to rate several sets of lectures in various subject areas and calculated the correlation between the ratings predicted by the model and those reported by the participants. Assuming participant P2 gave equal ratings to the Science and Literature lectures, the data in the graph indicate the model predicted that ______

[PLACEHOLDER: Bar chart titled 'Correlation between Model-Predicted and Student-Reported Engagement Ratings, by Subject Area.' X-axis shows two subject areas: Science and Literature. Y-axis shows Correlation from 0 to 0.5. Three bars per subject area represent participants P2 (light gray), P5 (white), and P6 (dark gray). For Science: P2 ~0.18, P5 ~0.38, P6 ~0.22. For Literature: P2 ~0.24, P5 ~0.44, P6 ~0.28.]

A) P2's ratings for Science and Literature lectures would differ from one another.
B) P2 would derive less engagement from Science lectures than from Literature lectures.
C) P2's rating for Science and Literature lectures would be identical.
D) P2 would derive more engagement from Science lectures than from Literature lectures.

Question 13

Skill: Inferences / Logical Completion | Difficulty: Medium

To understand listener loyalty among subscribers to specific podcast platforms, Dr. Carlos Vega and colleagues conducted a survey using 38 statements as proxy indicators of subscriber satisfaction with their listening experience. The statements were categorized by topic—e.g., statement 12, "I recommend this platform to friends," was categorized as advocacy behavior—and respondents, all of whom were from Sweden (which is characterized as having a highly saturated podcast market), rated the importance of each statement to their experience. Researchers found that respondents placed high importance on episode variety and personalized recommendations but low importance on app aesthetics, but the researchers cautioned against applying the findings to subscribers generally, suggesting that ______

A) additional research is needed with participants from countries of varying levels of podcast market saturation to determine whether episode variety and recommendations are more important than app aesthetics to subscribers broadly.
B) subscribers in countries with less saturated podcast markets than that of Sweden are likely to regard app aesthetics as relatively more important than episode variety and personalized recommendations.
C) podcast platforms that operate in a variety of countries are more likely to retain subscribers if they invest in episode variety and allow subscribers to customize recommendations than if they redesign their app interfaces.
D) the greater importance assigned to episode variety and recommendations than to app aesthetics may not be observed if the survey were to be given to a larger group of Swedish participants.

Question 14

Skill: Inferences / Logical Completion | Difficulty: Medium

Dr. Hannah Eriksson et al. tracked humidity-driven changes in the population of Varroa destructor (a parasitic mite that requires two host stages throughout its life cycle), Tropilaelaps mercedesae (a directly reproducing mite, which requires only one host environment), and 40 other arthropod taxa found in fifteen apiary systems. Arthropods with complex life cycles are typically buffered from humidity stress by their host organisms, whereas directly reproducing arthropods are exposed to external moisture conditions. However, Eriksson et al. found that two-host arthropod abundance decreased as humidity levels fluctuated, whereas directly reproducing arthropod abundance was largely stable, suggesting that ______

A) Arthropods primarily transmitted through host organisms were less dependent on host species adversely affected by humidity changes than were arthropods that use other reproductive strategies.
B) As the number of host species involved in an arthropod's life cycle increases, the arthropod is better protected against humidity fluctuations.
C) Directly reproducing arthropods identified in the study were more likely to use reproductive strategies that shield them from humidity changes than were two-host arthropods.
D) Any advantages that the life-cycle strategy used by two-host arthropods may have conferred did not fully offset the negative effects of other humidity-driven factors on their abundance.

Question 15

Skill: Inferences / Logical Completion | Difficulty: Medium

Prolonged exposure to urban noise pollution (sound from human sources like traffic or construction) can affect wildlife, as Dr. Mei Lin and colleagues found in a 2018 study of city-dwelling songbirds. Researchers conducted a meta-analysis of studies of how noise pollution affects diurnal animals and found that, for every study, relevant traits or behaviors of the animals were observably different between the noise-exposed group and the otherwise similar but unexposed group. Although, on average, studies of birds showed larger differences than studies of mammals did, for every class of animals examined, there were individual studies showing differences well above the average for birds. Therefore, the results of the meta-analysis suggest that ______

A) some studies of mammals found larger effects of exposure to urban noise pollution than some studies of birds did.
B) the studies in the meta-analysis that examined birds were more likely than those that examined mammals to specify whether the observed effects were harmful.
C) the differences that studies attribute to exposure to urban noise pollution are likely to be more pronounced for mammals than they are for birds.
D) the difference found in the study conducted by Dr. Mei Lin and colleagues was likely larger than the average difference for studies of songbirds included in the meta-analysis.

Question 16

Skill: Standard English Conventions | Difficulty: Easy

The Mughal Empire, which dominated the Indian subcontinent from around 1526 CE to 1857 CE, prospered from trade with neighboring civilizations: its merchants exported cotton, indigo, and silk and ______ other luxury goods, such as gemstones, ivory, and aromatic oils, which were then sold in regional marketplaces.

A) to import
B) imported
C) importing
D) having imported

Question 17

Skill: Standard English Conventions | Difficulty: Easy

In her work, M■ori-heritage artist Lisa Reihana uses video, projection, and other digital mediums to explore Indigenous identity and Pacific landscapes. The Auckland Art Gallery counts pieces by Reihana in ______ extensive collection of New Zealand art.

A) their
B) it's
C) his
D) its

Question 18

Skill: Standard English Conventions | Difficulty: Medium

On February 18, 2021, NASA's Perseverance rover detected a remarkable and previously unknown geological feature on Mars, located in the Jezero Crater, 292 million miles from ______ observation of a vast olivine-rich deposit, an unusually smooth mineral formation that typically results from sustained volcanic activity on a planet's surface.

A) Earth. The
B) Earth, it was the
C) Earth: the
D) Earth, the

Question 19

Skill: Standard English Conventions | Difficulty: Medium

Between late November and mid-January each year, debris shed by asteroid 3200 Phaethon enters Earth's atmosphere at a velocity of 35 kilometers per second. Relatively few of these particles are likely to reach Earth's ______ since the meteoroids' friction with the atmosphere causes the vast majority to disintegrate in the upper atmosphere, producing the Geminid meteor shower.

A) surface; however,
B) surface, however,
C) surface. However,
D) surface, however;

Question 20

Skill: Standard English Conventions | Difficulty: Medium

In a titration procedure, the quantity known as molarity helps determine the concentration of the analyte in the solution. The aqueous ______ have molarities of 0.10 and 0.25 mol/L, respectively.

A) reagents, hydrochloric acid and sodium hydroxide
B) reagents, hydrochloric acid and sodium hydroxide,
C) reagents hydrochloric acid and sodium hydroxide
D) reagents hydrochloric acid and sodium hydroxide,

Question 21

Skill: Standard English Conventions | Difficulty: Medium

The philosophy of the Reggio Emilia approach is grounded in several foundational ______ by Italian educator Loris Malaguzzi, who sought to honor children's natural curiosity, the approach emphasizes collaborative exploration and project-based learning and introduces students as young as three to artistic expression through activities such as painting and sculpting.

A) principles developed
B) principles, developed
C) principles. Developed
D) principles that, developed

Question 22

Skill: Transitions | Difficulty: Easy

Clara Schumann and Fanny Mendelssohn have both been posthumously recognized by the Royal Philharmonic Society, though for different musical contributions. Schumann was honored for her virtuoso piano performances; Mendelssohn, ______ was recognized for her compositions for chamber ensembles and choral groups.

A) as a result,
B) for instance,
C) in other words,
D) by contrast,

Question 23

Skill: Transitions | Difficulty: Easy

In a 2020 analysis of 200 terms, researchers found a broad pattern of frequency-dependent change: words used less often in everyday speech saw a faster rate of phonological drift—______ the rate at which a word's pronunciation will shift over time. Adjectives (e.g., "gentle") saw the largest effect; prepositions (e.g., "toward"), meanwhile, saw the smallest.

A) likewise,
B) that is,
C) for example,
D) in addition,

Question 24

Skill: Transitions | Difficulty: Medium

At previous Olympic Games, in contrast to events showcasing internationally recognized sports, regional martial arts were presented as limited-run demonstration events. ______ while the globally practiced sport of judo became a permanent Olympic event after its initial demonstration, kabaddi, an Indian contact sport popular mainly in South Asia, only appeared as a demonstration at the 1936 Berlin Olympics.

A) Moreover,
B) Accordingly,
C) Granted,
D) However,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

• Composer Heitor Villa-Lobos was a Brazilian musician and citizen of Rio de Janeiro.

• His compositions blend European classical forms with elements of traditional Brazilian folk music.

• Villa-Lobos's orchestral piece Bachianas Brasileiras No. 5 features a soprano voice singing a wordless melody.

• Villa-Lobos's orchestral piece Chôros No. 10 incorporates a cuíca, a traditional Brazilian friction drum.

The student wants to emphasize a difference between the two compositions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) While both compositions integrate Brazilian musical elements, Bachianas Brasileiras No. 5 does so by featuring a soprano voice and Chôros No. 10 does so by incorporating a cuíca.
B) Bachianas Brasileiras No. 5 and Chôros No. 10 are two compositions written by Villa-Lobos, a Brazilian composer.
C) In his two compositions Bachianas Brasileiras No. 5 and Chôros No. 10, Villa-Lobos blends elements of traditional Brazilian folk music, such as vocal melodies or percussion, with European classical forms.
D) Villa-Lobos has different approaches to blending European classical music with elements of Brazilian tradition, such as using voices and drums in his compositions.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

• A glacial lake is a body of water formed by the melting and retreating of a glacier.

• A glacial lake forms when meltwater fills a depression carved by the glacier's movement, creating a basin.

• Glacial lakes will gradually silt up and become meadows over thousands of years.

• A glacier with a high rate of retreat loses mass quickly, and a glacier with a low rate of retreat loses mass slowly.

• The Columbia Glacier in Alaska has a high rate of retreat.

The student wants to define the term "high-retreat glacier." Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) A high-retreat glacier is one that loses mass quickly, forming depressions that can fill with meltwater.
B) A high-retreat glacier has depressions called basins that will gradually fill with silt and become meadows.
C) High retreat is caused by the formation of glacial lakes from glacier meltwater over time.
D) Over time, the rapid mass loss of the high-retreat Columbia Glacier will create depressions and fill them with meltwater.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

• Saturn has at least 146 known moons, including Titan and Enceladus.

• Titan completes an orbit of Saturn in 15.95 Earth days on average.

• Enceladus completes an orbit of Saturn in 1.37 Earth days on average.

• German astronomer Johannes Kepler (1571–1630) described the orbits of planets as elliptical in shape rather than circular.

• Kepler published three laws of planetary motion to describe the orbits of planets around the Sun.

• Kepler's laws also apply to the elliptical orbits of natural satellites (e.g., moons).

The student wants to contrast the orbits of the two moons. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Titan's orbit of Saturn takes, on average, 15.95 Earth days, while Enceladus's orbit of Saturn takes only 1.37 Earth days on average.
B) Kepler's three laws of planetary motion describe the orbits of planets and natural satellites, including those of Titan and Enceladus.
C) Titan's orbit of Saturn is elliptical in shape and takes, on average, 15.95 Earth days, while Enceladus's orbit of Saturn is also elliptical and takes, on average, 1.37 Earth days.
D) Saturn's moons, specifically Titan and Enceladus, orbit their planet in the same way the planets orbit the Sun.

SECTION 1: READING AND WRITING

Module 2 — Harder Path (27 Questions)

Question 1

Skill: Vocabulary in Context | Difficulty: Medium

Although the architectural heritage of the Silk Road is remarkably diverse in form and material, it is also marked by a striking ______: scholars have identified design motifs in caravanserais from Uzbekistan that closely echo those found in steppe settlements as far east as the Gansu Corridor in China.

A) dissonance
B) coherence
C) modesty
D) ambiguity

Question 2

Skill: Vocabulary in Context | Difficulty: Hard

When evaluating the symphonies of Dmitri Shostakovich, musicologists frequently describe his Fifth Symphony as the quintessential example of state-sanctioned Soviet Realism. The basis for this characterization may seem _______, but categorizing the Fifth Symphony solely within that framework risks ignoring the subversive irony embedded in its triumphal finale, an element that resists easy classification within any single ideological program.

A) tenuous
B) provocative
C) incontrovertible
D) paradoxical

Question 3

Skill: Vocabulary in Context | Difficulty: Hard

The celebrated frescoes adorning the Sistine Chapel ceiling in Vatican City demonstrate Michelangelo's unparalleled contribution to Renaissance art, a legacy that is ______ throughout Western artistic tradition—students studying Delacroix's murals in the Palais Bourbon, for instance, can readily discern how their compositional dynamism draws inspiration from Michelangelo's figural arrangements.

A) authenticated
B) conspicuous
C) debated
D) attenuated

Question 4

Skill: Vocabulary in Context | Difficulty: Hard

In quantum chromodynamics, when physicists observe the behavior of quarks confined within a hadron, the strong nuclear force between the quarks is assigned a provisional coupling constant based on the energy scale and momentum transfer of the interaction. After the coupling has been verified through multiple independent measurements, the research group may propose a refined value. Under current conventions of the Particle Data Group, it is ______ for experimentalists to adjust the coupling constant based on lattice QCD calculations, but they are not compelled to do so; the final determination rests with the review committee.

A) permissible
B) unwarranted
C) selective
D) compulsory

Question 5

Skill: Text Structure and Purpose | Difficulty: Hard

The following text is from Fyodor Dostoevsky's 1880 novel *The Brothers Karamazov* (translated by Constance Garnett in 1912). The narrator is describing the elder monk Father Zossima receiving visitors at a monastery.

A young officer in a new uniform entered at that moment and seated himself on the bench nearest the icon screen. He regarded the assembled pilgrims with the weary impatience of a man accustomed to giving orders in barracks, as if to communicate, "I have conversed with bishops and metropolitans across the empire, and you are all fortunate that I have consented to enter this humble cell." There are visitors who approach monasteries as inspectors rather than supplicants, and it is well established that such persons rarely find peace in any sacred place.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It conveys the eagerness some monastery visitors feel to be noticed by the monks.
B) It emphasizes the young officer's desire to become a religious authority.
C) It helps explain why some military officers avoid monasteries altogether.
D) It illustrates the young officer's condescending attitude toward the other people in the room.

Question 6

Skill: Text Structure and Purpose | Difficulty: Hard

Beloved, first published in 1987, is a novel by Toni Morrison that draws on historical accounts of slavery in the antebellum United States. Like much of Morrison's fiction, *Beloved* employs a non-linear narrative structure, but the subsequent novel *Jazz* extends and deepens many of the same structural experiments. Thus, the fragmentation of the narrative is only disorienting if one treats the novels as isolated texts—<u>Morrison's trilogy is best understood as a single meditation on African American historical memory, comparable to Faulkner's interlinked Yoknapatawpha narratives</u>.

A) It demonstrates that readers who enjoy *Jazz* will inevitably prefer it to *Beloved*.
B) It argues that Morrison's trilogy merits the same canonical status as Faulkner's Yoknapatawpha novels.
C) It offers a framework that helps explain why a particular narrative feature of *Beloved* should not necessarily be regarded as a deficiency.
D) It presents evidence that *Beloved* and *Jazz* are Morrison's most critically discussed novels.

Question 7

Skill: Text Structure and Purpose | Difficulty: Hard

Argentinean novelist Jorge Luis Borges's prolific career, which spanned the 1920s to the 1980s, evolved through distinct literary phases. As epitomized by his early poetry collection *Fervor de Buenos Aires*, many of Borges's initial works celebrated the landscapes and mythology of his native city. Immersion in the philosophical traditions of Berkeley, Schopenhauer, and Spinoza later sparked an engagement with metaphysical fiction—stories whose emphasis on paradox, infinite regress, and epistemological uncertainty and whose departure from purely lyrical writing are readily apparent in Borges's collection *Ficciones*.

A) It presents biographical information that accounts for the transition in Borges's career that is discussed in the text.
B) It notes why the two literary works described in the text constitute major departures from Borges's typical style.
C) It lends support to an argument about the lyrical aesthetic that is made in the previous sentence.
D) It explains why the literary work discussed later in the sentence was highly regarded by Borges's contemporaries.

Question 8

Skill: Central Ideas and Details | Difficulty: Hard

Ghanaian textile artist El Anatsui transforms gallery spaces using his distinctive method of assembling thousands of discarded bottle caps and aluminum fragments into shimmering, monumental tapestries. In some installations, Anatsui focuses on commemorating the transatlantic slave trade, as he does in his wall sculpture inspired by the trade routes between West Africa and the Americas. However, in other installations, Anatsui celebrates the resourcefulness of everyday market vendors, as he does in his radiant depiction of a cloth merchant's stall in Nsukka. Anatsui views both of these approaches as ways of representing the adaptability and interconnectedness of all communities.

A) Anatsui began his artistic career by commemorating historical atrocities and then transitioned to depicting ordinary community members instead.
B) Anatsui's portrayal of a cloth merchant emphasizes communal ingenuity more than it does individual suffering.
C) Anatsui's installations honor both historically significant events and everyday examples of human resourcefulness and connection.
D) Anatsui's depiction of the slave trade receives more critical attention than his depiction of the cloth merchant does.

Question 9

Skill: Central Ideas and Details | Difficulty: Hard

In their systematic review of research on employer-sponsored wellness programs (initiatives that offer incentives such as gym memberships, health screenings, and smoking cessation support to encourage healthier employee behavior), Dr. Ananya Sharma et al. confirm that such programs, though they can reduce short-term absenteeism, may not translate into measurable decreases in long-term healthcare costs. This phenomenon can be explained by the concept of moral hazard as it is described in Kenneth Arrow's foundational work on health economics. In this case, employees who receive free or subsidized wellness benefits may reduce their own preventive efforts outside the program, leaving overall health expenditures largely unchanged.

A) Arrow's concept of moral hazard provides a framework for explaining the finding by Sharma et al. that wellness programs may not reduce long-term healthcare costs because employees offset program benefits with reduced personal effort.
B) The systematic review by Sharma et al. reveals that higher participation in wellness programs is linked to lower long-term productivity among employees.
C) Although Arrow's health economics framework suggests otherwise, employer wellness programs can succeed in reducing healthcare costs, provided they balance incentives and personal accountability.
D) Research by Sharma et al. supports Arrow's conclusion that because employees primarily focus on subsidized benefits, it is impossible for wellness programs to reduce healthcare expenditures over time.

Question 10

Skill: Command of Evidence (Strengthen/Weaken) | Difficulty: Hard

Dr. Lucia Ferretti and Dr. Hiroshi Nakamura investigated the role of mycorrhizal networks (underground fungal connections between tree roots) in nutrient sharing among forest tree species. They found that during controlled greenhouse experiments, 55% of the tested tree pairs showed measurable nutrient transfer through shared mycorrhizal networks, with transfer rates varying from rapid in birch-fir pairs to negligible in oak-pine pairs. A second research team has claimed that trees use these mycorrhizal networks primarily to support the growth of their own offspring (kin selection).

A) Among the tree pairs that showed nutrient transfer, no individuals transferred nutrients to unrelated trees, but the vast majority transferred nutrients to genetically related seedlings.
B) Most of the tree pairs that showed high nutrient transfer in mycorrhizal experiments were composed of unrelated species that do not share recent common ancestors.
C) Tree pairs that were found to transfer nutrients in mycorrhizal experiments were all from the same genus and were planted in identical soil conditions.
D) The nutrients transferred through mycorrhizal networks had molecular signatures indicating they originated from decomposing leaf litter rather than from the connected living trees.

Question 11

Skill: Command of Evidence (Research Design) | Difficulty: Hard

Patagonia's expansion from outdoor clothing into food products in 2012 through its Patagonia Provisions line is a noteworthy instance of values-driven brand extension—the company leveraged its brand identity as an environmental advocate to enter a product category where it had not previously competed. An outstanding question is whether perceived mission alignment predicts consumers' willingness to pay a premium for values-driven brand extensions. To answer this question, Dr. Kofi Mensah et al. identified 20 extended-brand pairs (e.g., the same brand of outdoor apparel and packaged food) in 40 weeks of purchases by approximately 45,000 households and, for each pair, calculated the price premium consumers paid for the extended-brand product relative to comparable unbranded alternatives.

A) Have a representative sample of the households rate the degree of mission alignment between the original brand and its extension in each pair, then determine how, if at all, those ratings correlate with the price premium the team calculated for each pair.
B) Poll a representative sample of the households to determine how much each household spent annually on environmental causes, then determine how, if at all, that spending correlates with the average cost of each product in the pairs.
C) Have a representative sample of the households rate the quality of one product in each extended-brand pair relative to comparable unbranded products, then determine how, if at all, those ratings correlate with the price premium that the team calculated for each pair.
D) Poll a representative sample of the households to determine their overall familiarity with each brand in the extended-brand pairs, then determine how, if at all, that familiarity correlates with the frequency with which a different group of households purchased at least one product of that brand.

Question 12

Skill: Command of Quantitative Evidence | Difficulty: Hard

Neuroscientist Dr. Priya Chakraborty and colleagues developed a computational model to predict the intensity of a patient's pain response to a particular stimulus on a scale from 1 (no pain) to 10 (extreme pain). They then recruited participants to use the same scale to rate several sets of stimuli in various modalities and calculated the correlation between the ratings predicted by the model and those reported by the participants. Assuming participant P3 reported identical pain scores for the thermal and pressure stimuli, the data in the graph indicate the model predicted that ______

[PLACEHOLDER: Bar chart titled 'Correlation between Model-Predicted and Patient-Reported Pain Ratings, by Stimulus Modality.' X-axis shows two modalities: Thermal and Pressure. Y-axis shows Correlation from 0 to 0.6. Three bars per modality represent participants P3 (light gray), P7 (white), and P9 (dark gray). For Thermal: P3 ~0.22, P7 ~0.48, P9 ~0.30. For Pressure: P3 ~0.32, P7 ~0.52, P9 ~0.36.]

A) P3's pain ratings for thermal and pressure stimuli would diverge from one another.
B) P3 would experience less pain from thermal stimuli than from pressure stimuli.
C) P3's pain rating for thermal and pressure stimuli would be identical.
D) P3 would experience more pain from thermal stimuli than from pressure stimuli.

Question 13

Skill: Inferences / Logical Completion | Difficulty: Hard

To understand viewer loyalty among subscribers to specific art-house cinema streaming platforms, Dr. Ingrid Solberg and colleagues conducted a survey using 50 statements as proxy indicators of subscriber satisfaction with their viewing experience. The statements were categorized by topic—e.g., statement 23, "I actively seek out films from this platform's catalog," was categorized as exploratory engagement—and respondents, all of whom were from France (which is characterized as having a highly developed independent film culture), rated the importance of each statement to their experience. Researchers found that respondents placed high importance on curatorial distinctiveness and director-specific collections but low importance on platform loading speed, but the researchers cautioned against applying the findings to subscribers generally, suggesting that ______

A) additional research is needed with participants from countries of varying levels of independent film culture to determine whether curatorial distinctiveness and director collections are more important than loading speed to subscribers broadly.
B) subscribers in countries with less developed independent film cultures than that of France are likely to regard loading speed as relatively more important than curatorial distinctiveness and director-specific collections.
C) art-house streaming platforms that operate in a variety of countries are more likely to retain subscribers if they invest in curatorial distinctiveness than if they improve loading speed.
D) the greater importance assigned to curatorial distinctiveness and director collections than to loading speed may not be observed if the survey were to be given to a larger group of French participants.

Question 14

Skill: Inferences / Logical Completion | Difficulty: Hard

Dr. Sofia Lindgren et al. tracked salinity-driven changes in the population of Schistosoma mansoni (a parasitic flatworm that requires two host organisms throughout its life cycle), Dugesia japonica (a directly reproducing flatworm, which requires only one aquatic environment), and 50 other invertebrate taxa found in fourteen estuarine systems. Invertebrates with complex life cycles are typically buffered from salinity stress by their host organisms, whereas directly reproducing invertebrates are exposed to external salinity conditions. However, Lindgren et al. found that two-host invertebrate abundance decreased as salinity levels fluctuated, whereas directly reproducing invertebrate abundance was largely stable, suggesting that ______

A) Invertebrates primarily transmitted through host organisms were less dependent on host species adversely affected by salinity changes than were invertebrates that use other reproductive strategies.
B) As the number of host species involved in an invertebrate's life cycle increases, the invertebrate is better protected against salinity fluctuations.
C) Directly reproducing invertebrates identified in the study were more likely to use reproductive strategies that shield them from salinity changes than were two-host invertebrates.
D) Any advantages that the life-cycle strategy used by two-host invertebrates may have conferred did not fully offset the negative effects of other salinity-driven factors on their abundance.

Question 15

Skill: Inferences / Logical Completion | Difficulty: Hard

Chronic exposure to microplastic contamination (particles smaller than 5 mm from degraded plastic waste) can affect marine organisms, as Dr. Anders Björk and colleagues found in a 2020 study of deep-water coral colonies. Researchers conducted a meta-analysis of studies of how microplastic exposure affects sessile marine organisms and found that, for every study, relevant physiological traits of the organisms were measurably different between the contamination-exposed group and the otherwise similar but unexposed group. Although, on average, studies of corals showed larger differences than studies of sponges did, for every phylum of organisms examined, there were individual studies showing differences well above the average for corals. Therefore, the results of the meta-analysis suggest that ______

A) some studies of sponges found larger effects of exposure to microplastic contamination than some studies of corals did.
B) the studies in the meta-analysis that examined corals were more likely than those that examined sponges to specify whether the observed effects were irreversible.
C) the differences that studies attribute to microplastic exposure are likely to be more pronounced for sponges than they are for corals.
D) the difference found in the study conducted by Dr. Anders Björk and colleagues was likely larger than the average difference for studies of corals included in the meta-analysis.

Question 16

Skill: Standard English Conventions | Difficulty: Hard

The Byzantine Empire, which dominated the eastern Mediterranean from approximately 330 CE to 1453 CE, flourished through commerce with neighboring civilizations: its merchants exported silk, dyes, and metalwork and ______ other luxury commodities, such as amber, furs, and enslaved laborers, which were then distributed through imperial trade networks.

A) to acquire
B) acquired
C) acquiring
D) having acquired

Question 17

Skill: Standard English Conventions | Difficulty: Hard

In his scholarship, Lakota-heritage historian Nick Estes uses archival documents, oral histories, and other primary sources to examine Indigenous sovereignty and environmental justice. The Library of Congress counts volumes by Estes in ______ growing collection of Native American studies materials.

A) their
B) it's
C) his
D) its

Question 18

Skill: Standard English Conventions | Difficulty: Hard

On November 26, 2018, NASA's InSight lander detected a remarkable and previously unrecorded seismic event on Mars, originating in the Cerberus Fossae region, 225 million miles from ______ registration of a sustained low-frequency tremor, an exceptionally faint oscillation that typically results from deep tectonic fracturing of a planet's lithosphere.

A) Earth. The
B) Earth, it was the
C) Earth: the
D) Earth, the

Question 19

Skill: Standard English Conventions | Difficulty: Hard

Between early December and late February each year, debris shed by comet 109P/Swift-Tuttle enters Earth's atmosphere at a velocity of 59 kilometers per second. Relatively few of these particles are likely to reach Earth's ______ since the meteoroids' friction with the atmosphere causes the vast majority to vaporize in the thermosphere, producing the Perseid meteor shower.

A) surface; however,
B) surface, however,
C) surface. However,
D) surface, however;

Question 20

Skill: Standard English Conventions | Difficulty: Hard

In a spectroscopic analysis, the quantity known as wavelength helps differentiate between the emission lines of the elements. The gaseous ______ have characteristic emission wavelengths of 589.0 and 656.3 nanometers, respectively.

A) elements, sodium and hydrogen
B) elements, sodium and hydrogen,
C) elements sodium and hydrogen
D) elements sodium and hydrogen,

Question 21

Skill: Standard English Conventions | Difficulty: Hard

The pedagogy of the Waldorf method is anchored in several core ______ by Austrian philosopher Rudolf Steiner, who sought to integrate artistic expression into every academic subject, the method emphasizes imaginative play and oral storytelling and introduces students as young as six to handwork exercises such as knitting and woodcarving.

A) tenets formulated
B) tenets, formulated
C) tenets. Formulated
D) tenets that, formulated

Question 22

Skill: Transitions | Difficulty: Hard

Dorothea Lange and Walker Evans have both been honored with retrospective exhibitions at the Museum of Modern Art, though for different photographic legacies. Lange was celebrated for her documentation of migrant farmworkers during the Depression; Evans, ______ was recognized for his architectural studies of vernacular American buildings.

A) as a result,
B) for instance,
C) in other words,
D) by contrast,

Question 23

Skill: Transitions | Difficulty: Hard

In a 2023 analysis of 180 lexical items, linguists found a broad pattern of register-dependent change: words used less frequently in formal academic prose saw a faster rate of connotative shift—______ the rate at which a word's evaluative associations will evolve over time. Abstract nouns (e.g., "honor") saw the largest effect; concrete nouns (e.g., "stone"), meanwhile, saw the smallest.

A) likewise,
B) that is,
C) for example,
D) in addition,

Question 24

Skill: Transitions | Difficulty: Hard

At previous World Expositions, in contrast to pavilions showcasing internationally recognized architectural innovations, regional vernacular building techniques were exhibited as temporary demonstration structures. ______ while the internationally celebrated Crystal Palace design became a permanent influence on exhibition architecture after its debut at the 1851 London Exposition, the traditional Ainu chise dwelling, a thatched-roof structure popular mainly in Hokkaido, only appeared at the 1970 Osaka Expo.

A) Moreover,
B) Accordingly,
C) Granted,
D) However,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• Architect Tadao Ando was a Japanese designer and resident of Osaka.

• His buildings blend minimalist concrete forms with elements of traditional Japanese spatial philosophy.

• Ando's Church of the Light features a cruciform aperture that allows a single beam of sunlight to penetrate the interior.

• Ando's Water Temple features an oval lotus pond through which visitors descend into a subterranean worship hall.

The student wants to emphasize a difference between the two buildings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) While both buildings integrate natural elements into their design, the Church of the Light does so through a cruciform aperture admitting sunlight and the Water Temple does so through a lotus pond serving as an entrance.
B) The Church of the Light and the Water Temple are two buildings designed by Ando, a Japanese architect.
C) In his two buildings, the Church of the Light and the Water Temple, Ando blends elements of Japanese spatial philosophy, such as light or water, with minimalist concrete forms.
D) Ando has different approaches to blending minimalism with Japanese spatial philosophy, such as using natural light and water features in his buildings.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• A caldera is a large, basin-shaped volcanic depression formed when a volcano erupts and its magma chamber collapses.

• A caldera forms when the eruption empties the underlying magma reservoir, causing the surface above to subside.

• Calderas will gradually fill with water and become crater lakes over geological time.

• A volcano with high eruptive frequency has many recorded eruptions, and a volcano with low eruptive frequency has few.

• Mount Aso in Japan has high eruptive frequency.

The student wants to define the term "high-eruptive-frequency volcano." Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) A high-eruptive-frequency volcano is one that has many recorded eruptions, often producing large basin-shaped depressions called calderas.
B) A high-eruptive-frequency volcano has basin-shaped depressions called calderas that will gradually fill with water and become crater lakes.
C) High eruptive frequency is caused by the repeated collapse of magma chambers over geological time.
D) Over time, the many eruptions of the high-eruptive-frequency Mount Aso will produce additional calderas and potentially new crater lakes.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• Neptune has at least 16 known moons, including Triton and Nereid.

• Triton completes an orbit of Neptune in 5.88 Earth days on average.

• Nereid completes an orbit of Neptune in 360.13 Earth days on average.

• Triton orbits Neptune in a retrograde direction (opposite to Neptune's rotation).

• Nereid orbits Neptune in a prograde direction (same as Neptune's rotation).

• Kepler's laws apply to the elliptical orbits of natural satellites.

The student wants to contrast the orbits of the two moons. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Triton orbits Neptune in a retrograde direction and completes an orbit in 5.88 Earth days, while Nereid orbits in a prograde direction and takes 360.13 Earth days to complete an orbit.
B) Kepler's three laws of planetary motion describe the orbits of planets and natural satellites, including those of Triton and Nereid.
C) Triton completes an orbit of Neptune much more quickly than Nereid, which takes over 360 Earth days.
D) Neptune's moons, specifically Triton and Nereid, orbit their planet in the same way the planets orbit the Sun.

SECTION 2: MATH

Module 1 (22 Questions — All Students)

Question 1

Skill: Functions / Exponential vs. Linear | Difficulty: Easy

The table shows selected values from the function h.

[PLACEHOLDER: Table with two columns: x and h(x). Rows: x = –1, h(x) = 1/37; x = 0, h(x) = 1; x = 1, h(x) = 37; x = 2, h(x) = 1,369. Title: 'Which of the following is the best description of function h?']

A) Increasing linear
B) Decreasing linear
C) Increasing exponential
D) Decreasing exponential

Question 2

Skill: Linear Functions | Difficulty: Easy

The function p is defined by $p(x) = -7x + 42$. What is the value of $p(x)$ when $x = -3$?

A) 63
B) 56
C) 21
D) 42

Question 3

Skill: Equations | Difficulty: Easy

$x^2 = (36)(36)$. What is the positive solution to the given equation?

Question 4

Skill: Statistics (Mean/Median) | Difficulty: Easy

Data set C and data set D each have 11 values. Which of the following statements best compares the median of data set C and the median of data set D?

[PLACEHOLDER: Data set C: 2, 2, 2, 5, 5, 8, 11, 11, 14, 14, 14. Data set D: 2, 5, 5, 8, 8, 8, 8, 11, 11, 14, 14.]

A) The median of data set C is greater than the median of data set D.
B) The median of data set C is less than the median of data set D.
C) The medians of data sets C and D are equal.
D) There is not enough information to compare the medians.

Question 5

Skill: Functions / Polynomial | Difficulty: Easy

The function g is defined by $g(x) = x^3 + 27$. What is the value of $g(2)$?

Question 6

Skill: Exponential Functions / Transformations | Difficulty: Easy

In the xy-plane, the graph of the equation $y = 5^x$ is translated 3 units down. What is an equation of the resulting graph?

- A) $y = 5^x + 3$
- B) $y = 5^x - 3$
- C) $y = 5^{(x+3)}$

Question 7

Skill: Systems of Equations | Difficulty: Medium

$y = 3x + 8$
$y = (x + 1)^2$

A solution to the given system of equations is (x, y). What is the value of x^2?

Question 8

Skill: Interpreting Functions in Context | Difficulty: Easy

$f(x) = -4x^2 + 56x$

The function f models the height above ground level, in meters, of a certain rocket x seconds after it is launched. What is the best interpretation of $f(3) = 132$ in this context?

- A) 132 seconds after the rocket is launched, its estimated height is 3 meters above ground level.
- B) The rocket reaches an estimated maximum height of 132 meters above ground level.
- C) 3 seconds after the rocket is launched, its estimated height is 132 meters above ground level.
- D) Approximately 3 seconds after launching, the rocket returns to ground level.

Question 9

Skill: Linear Equations in Context | Difficulty: Easy

A community theater sold tickets to raise money. The theater raised $10 from each student ticket sold and raised $15 from each senior ticket sold. The theater raised a total of $370 from the sale of these tickets. The equation $10x + 15y = 370$ represents this situation. Which of the following is the best interpretation of y in this context?

- A) The number of senior tickets sold
- B) The number of student tickets sold
- C) The price, in dollars, of each senior ticket
- D) The price, in dollars, of each student ticket

Question 10

Skill: Systems of Equations / Mixture | Difficulty: Medium

A chemist mixed x liters of a 5% acid solution with y liters of a 12% acid solution to produce an 8% acid solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)

- A) $0.05x + 0.12y = 8(x + y)$
- B) $0.05x + 0.12y = 0.08(x + y)$
- C) $0.5x + 1.2y = 8(x + y)$
- D) $0.5x + 1.2y = 0.8(x + y)$

Question 11

Skill: Trigonometry | Difficulty: Medium

In the right triangle shown, d is a constant. The value of cos x° = m/d, where m is a constant. What is the value of m?

[PLACEHOLDER: Figure: A right triangle with legs of length 18 and 24, hypotenuse d, and angle x° adjacent to the side of length 18. Note: Figure not drawn to scale.]

Question 12

Skill: Data Interpretation (Bar Graph) | Difficulty: Easy

The bar graph shows the frequency of each data value in a data set. What is the frequency of data value 3 in this data set?

[PLACEHOLDER: Bar graph showing the frequency of each data value in a data set. X-axis: Data value (1 through 6). Y-axis: Frequency (0 through 10). Bar heights: value 1 = 4, value 2 = 7, value 3 = 5, value 4 = 9, value 5 = 6, value 6 = 3.]

Question 13

Skill: Similar Triangles / Ratios | Difficulty: Medium

The table gives the perimeters of similar triangles GHI and JKL, where GH corresponds to JK. The length of GH is 15.

[PLACEHOLDER: Table: Perimeter of Triangle GHI = 45; Perimeter of Triangle JKL = 315. Question: What is the length of JK?]

- A) 5
- B) 15
- C) 60
- D) 105

Question 14

Skill: Geometry (Circles) | Difficulty: Easy

What is the radius of the circle in the xy-plane defined by $(x + 4)^2 + (y + 7)^2 = 169$?

Question 15

Skill: Linear Equations / Word Problem | Difficulty: Medium

A length of cable measuring 96 inches is cut into two pieces. One piece has a length of x inches, and the other piece has a length of y inches. The value of x is 14 more than 5 times the value of y. What is the value of x?

- A) 13.67
- B) 18
- C) 82.33
- D) 94

Question 16

Skill: Geometry (Area / Quadratics) | Difficulty: Medium

The area of a rectangle is 220 square centimeters. The length of the rectangle is 9 centimeters greater than the width of the rectangle. What is the width, in centimeters, of the rectangle?

- A) 11
- B) 13
- C) 20
- D) 22

Question 17

Skill: Inequalities / Word Problem | Difficulty: Medium

A freelance editor charges $35 per hour for the first 25 hours worked in a week. For each hour worked over 25 hours, the editor charges $55 per hour. What is the least number of whole hours that the editor must work in a week to earn at least $1,750?

- A) 40
- B) 41
- C) 15
- D) 14

Question 18

Skill: Systems of Equations (Graphical) | Difficulty: Medium

The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y). What is a possible value of x?

[PLACEHOLDER: Graph: A coordinate plane showing a horizontal line at y ≈ 6 and a downward-opening parabola with vertex near (6, 10). The two graphs intersect at approximately x = 3 and x = 9. Y-axis from –1 to 12, X-axis from 0 to 15.]

- A) 4
- B) 6
- C) 8.5
- D) 10

Question 19

Skill: Linear Equations (Number of Solutions) | Difficulty: Easy

How many solutions does the equation $11(x – 8) = –9(x + 14)$ have?

A) Exactly one
B) Exactly two
C) Infinitely many
D) Zero

Question 20

Skill: Systems of Equations | Difficulty: Hard

$14x – 11y = 29$
$9x + 13y = 47$

The solution to the given system of equations is $(x, y) = (p/59, w/826)$, where p and w are integers. What is the value of p?

Question 21

Skill: Linear Relationships | Difficulty: Hard

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. What is the x-coordinate of the x-intercept of the graph that represents this linear relationship in the xy-plane?

[PLACEHOLDER: Table: x = 2, y = –183/10; x = 4, y = –169/10; x = 7, y = –148/10.]

Question 22

Skill: Geometry (Similar Triangles) | Difficulty: Hard

In the figure, AC intersects DF at point E, and AD is parallel to FC. The lengths of DE, AE, and EF are 9, 10, and 12, respectively. What is the length of AC?

[PLACEHOLDER: Figure: Line segment AC intersects line segment DF at point E. AD is parallel to FC. The lengths of DE, AE, and EF are 9, 10, and 12, respectively.]

A) 262/12
B) 130/9
C) 238/9
D) 262/9

SECTION 2: MATH

Module 2 — Easier Path (22 Questions)

Question 1

Skill: Linear Equations | Difficulty: Easy

If $48 = 6 + 3(x - 5)$, what is the value of $3(x - 5)$?

A) 6
B) 30
C) 42
D) 48

Question 2

Skill: Exponents and Polynomials | Difficulty: Easy

Which expression is equivalent to $(x^2 + 4)^2 + (x - 3)(x + 3)$?

A) $x^4 + x^2 + 25$
B) $x^4 + 8x^2 + 16$
C) $x^4 + 9x^2 - 9$
D) $x^4 + 9x^2 + 7$

Question 3

Skill: Functions (Graph) | Difficulty: Easy

The graph of the polynomial function g in the xy-plane, where $y = g(x)$, passes through the point (4, 9). What is the value of $g(4)$?

A) 4
B) 9
C) 13
D) 36

Question 4

Skill: Exponential Functions | Difficulty: Easy

What is the y-intercept of the graph of $y = 5(2)^{x+3}$ in the xy-plane?

A) (0, 5)
B) (0, 10)
C) (0, 40)
D) (0, 320)

Question 5

Skill: Inequalities | Difficulty: Easy

$y > 2x - 5$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

[PLACEHOLDER: Four tables (A–D), each with three rows of x and y values as listed in the choices.]

A) x=3, y=0; x=5, y=4; x=8, y=10
B) x=3, y=4; x=5, y=8; x=8, y=14
C) x=3, y=1; x=5, y=5; x=8, y=11
D) x=3, y=2; x=5, y=6; x=8, y=12

Question 6

Skill: Linear Relationships | Difficulty: Easy

The table shows the linear relationship between the number of carriages, c, on a heritage railway train and the maximum number of passengers, p, that the train can carry. Which equation represents the linear relationship between c and p?

[PLACEHOLDER: Table: c = 3, p = 78; c = 5, p = 134; c = 10, p = 274.]

A) $28c - p = -6$
B) $28c - p = 6$
C) $28p - c = -6$
D) $28p - c = 6$

Question 7

Skill: Functions | Difficulty: Easy

The function k is defined by $k(x) = 4(5x - 6)$. What is the value of $k(7)$?

A) 29
B) 56
C) 116
D) 136

Question 8

Skill: Coordinate Geometry / Linear Equations | Difficulty: Easy

Line r in the xy-plane is defined by $y = x + 2/5$ and passes through the points (a, 0) and (0, b), where a and b are constants. What is the value of a?

Question 9

Skill: Quadratic Equations | Difficulty: Medium

$x^2 + 6x = 72$

What is one of the solutions to the given equation?

A) $3 + \sqrt{81}$
B) $–3 + \sqrt{81}$
C) $–3 + \sqrt{63}$
D) $3 + \sqrt{63}$

Question 10

Skill: Statistics (Standard Deviation) | Difficulty: Medium

The values in data sets R and S are shown in the table. The standard deviation of data set R is u, and the standard deviation of data set S is v. Which of the following statements about the standard deviations of the data sets is true?

[PLACEHOLDER: Data set R: 8, 8, 9, 9, 10, 11, 11, 12, 12, 12. Data set S: 3, 3, 4, 4, 5, 6, 6, 7, 7, 7.]

A) $u < v$
B) $u > v$
C) $u = v$
D) The relationship between u and v cannot be determined.

Question 11

Skill: Ratios and Proportions | Difficulty: Easy

The ratio 9 to x is equivalent to the ratio y to 4. Which equation represents x in terms of y?

A) $x = 9y/4$
B) $x = 4y/9$
C) $x = y/36$
D) $x = 36/y$

Question 12

Skill: Exponential Growth/Decay | Difficulty: Medium

For an online trivia game, 500 points are awarded for a correct answer if a question is answered in less than 5 seconds from the time the question is asked. Each 5 seconds after the question is asked, the number of points awarded for a correct answer decreases by 20% of the number of points awarded for a correct answer in the previous 5 seconds. Which function gives the number of points awarded for a correct answer x seconds from the time the question is asked, where x is a multiple of 5?

A) $f(x) = 500(0.80)^{x/5}$
B) $f(x) = 500(0.80)^{5x}$
C) $f(x) = 500(0.20)^{x/5}$
D) $f(x) = 500(0.20)^{5x}$

Question 13

Skill: Linear Equations / Word Problem | Difficulty: Easy

A workshop produces 9-inch, 6-inch, and 3-inch nails. During a certain shift, the number of 9-inch nails that the workshop produces is 4 times the number n of 6-inch nails, and the number of 3-inch nails is 15. During this shift, the workshop produces 75 nails total. Which equation represents this situation?

A) $9(4n) + 6n + 3(15) = 75$
B) $9n + 6n + 3n = 75$
C) $5n + 15 = 75$
D) $4n + 15 = 75$

Question 14

Skill: Coordinate Geometry (Circles / Tangent Lines) | Difficulty: Hard

A circle in the xy-plane has its center at (3, 6). Line n is tangent to this circle at the point (a, –1), where a is a constant. The slope of line n is 5/7. What is the value of a?

Question 15

Skill: Coordinate Geometry (Distance / Area) | Difficulty: Medium

The line segment shown in the xy-plane represents one of the legs of a right triangle. The area of this triangle is $25\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

[PLACEHOLDER: Graph: A line segment in the xy-plane from approximately (–4, 3) to (8, 6) representing one leg of a right triangle. The area of this triangle is $25\sqrt{13}$ square units.]

A) 10
B) 20
C) $5\sqrt{13}$
D) $10\sqrt{13}$

Question 16

Skill: Geometry (Triangle Congruence) | Difficulty: Medium

In triangle ABC and triangle DEF, AB and DE are each equal to 10 centimeters, and angles A and D each have measure 55°. Which additional piece of information is sufficient to prove that triangle ABC is congruent to triangle DEF?

A) The lengths of sides BC and EF are equal.
B) The lengths of sides AC and DF are equal.
C) The measures of angles B and E are equal.
D) No additional information is necessary to prove that the two triangles are congruent.

Question 17

Skill: Linear Functions (Graph Interpretation) | Difficulty: Medium

The graph of the linear function $y = f(x) - 12$ is shown. If c and d are positive constants, which equation could define f?

[PLACEHOLDER: Graph: A coordinate plane showing a line with positive slope, passing through approximately (–5, –8) and (5, 12). The y-intercept of the displayed line appears to be around (0, 2).]

A) $f(x) = d - cx$
B) $f(x) = -d + cx$
C) $f(x) = d + cx$
D) $f(x) = -d - cx$

Question 18

Skill: Probability | Difficulty: Medium

For 90 specimens, the table summarizes the distribution of mineral classification and crystal size. One of these specimens will be selected at random. What is the probability of selecting a specimen with a crystal size that is less than or equal to 20 mm, given that it is not classified as Quartz? (Express your answer as a decimal or fraction, not as a percent.)

[PLACEHOLDER: Table with columns: Classification, Less than 10 mm, 10 to 20 mm, Greater than 20 mm. Rows: Feldspar: 13, 8, 5; Quartz: 0, 10, 18; Mica: 9, 22, 5. Total specimens = 90.]

Question 19

Skill: Radical Equations | Difficulty: Hard

$\sqrt{(k - x)} = 32 - x$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of 4k?

Question 20

Skill: Factoring Polynomials | Difficulty: Medium

Which of the following expressions has a factor of $x + 3b$, where b is a positive integer constant?

A) $2x^2 + 9x + 18b$
B) $2x^2 + 13x + 18b$
C) $2x^2 + 15x + 18b$
D) $2x^2 + 21x + 18b$

Question 21

Skill: Quadratic Equations / Discriminant | Difficulty: Hard

$x(kx - 48) = -14$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

Question 22

Skill: Ratios and Percentages | Difficulty: Medium

The mass of object A is 250% of the mass of object B, and the mass of object A is 0.050% of the mass of object C. If the mass of object C is q% of the mass of object B, what is the value of q/1,000?

SECTION 2: MATH

Module 2 — Harder Path (22 Questions)

Question 1

Skill: Linear Equations | Difficulty: Medium

If $54 = 9 + 5(x – 3)$, what is the value of $5(x – 3)$?

A) 9
B) 35
C) 45
D) 54

Question 2

Skill: Exponents and Polynomials | Difficulty: Hard

Which expression is equivalent to $(x^2 + 7)^2 + (x – 5)(x + 5)$?

A) $x^4 + x^2 + 74$
B) $x^4 + 14x^2 + 49$
C) $x^4 + 15x^2 – 25$
D) $x^4 + 15x^2 + 24$

Question 3

Skill: Functions (Graph) | Difficulty: Medium

The graph of the polynomial function h in the xy-plane, where $y = h(x)$, passes through the point (–3, 17). What is the value of $h(–3)$?

A) –3
B) 17
C) 14
D) –51

Question 4

Skill: Exponential Functions | Difficulty: Hard

What is the y-intercept of the graph of $y = 6(4)^{x+3}$ in the xy-plane?

A) (0, 6)
B) (0, 24)
C) (0, 96)
D) (0, 384)

Question 5

Skill: Inequalities | Difficulty: Hard

$y > 4x - 7$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

[PLACEHOLDER: Four tables (A–D), each with three rows of x and y values as listed in the choices.]

A) $x=2$, $y=0$; $x=5$, $y=12$; $x=9$, $y=28$
B) $x=2$, $y=4$; $x=5$, $y=16$; $x=9$, $y=30$
C) $x=2$, $y=1$; $x=5$, $y=13$; $x=9$, $y=29$
D) $x=2$, $y=2$; $x=5$, $y=14$; $x=9$, $y=29$

Question 6

Skill: Linear Relationships | Difficulty: Hard

The table shows the linear relationship between the number of modules, m, in a solar panel array and the maximum power output, p (in watts), of the array. Which equation represents the linear relationship between m and p?

[PLACEHOLDER: Table: m = 2, p = 78; m = 4, p = 168; m = 8, p = 348.]

A) $45m - p = -12$
B) $45m - p = 12$
C) $45p - m = -12$
D) $45p - m = 12$

Question 7

Skill: Functions | Difficulty: Hard

The function j is defined by $j(x) = 5(3x - 11)$. What is the value of $j(9)$?

A) 16
B) 55
C) 80
D) 124

Question 8

Skill: Coordinate Geometry / Linear Equations | Difficulty: Hard

Line s in the xy-plane is defined by $y = x + 5/8$ and passes through the points $(a, 0)$ and $(0, b)$, where a and b are constants. What is the value of a?

Question 9

Skill: Quadratic Equations | Difficulty: Hard

$x^2 + 10x = 96$

What is one of the solutions to the given equation?

A) $5 + \sqrt{121}$
B) $-5 + \sqrt{121}$
C) $-5 + \sqrt{71}$
D) $5 + \sqrt{71}$

Question 10

Skill: Statistics (Standard Deviation) | Difficulty: Hard

The values in data sets T and U are shown in the table. The standard deviation of data set T is w, and the standard deviation of data set U is z. Which of the following statements about the standard deviations of the data sets is true?

[PLACEHOLDER: Data set T: 15, 15, 16, 16, 17, 18, 18, 19, 19, 19. Data set U: 6, 6, 7, 7, 8, 9, 9, 10, 10, 10.]

A) $w < z$
B) $w > z$
C) $w = z$
D) The relationship between w and z cannot be determined.

Question 11

Skill: Ratios and Proportions | Difficulty: Hard

The ratio 11 to x is equivalent to the ratio y to 8. Which equation represents x in terms of y?

A) $x = 11y/8$
B) $x = 8y/11$
C) $x = y/88$
D) $x = 88/y$

Question 12

Skill: Exponential Growth/Decay | Difficulty: Hard

For an arcade game, 800 points are awarded for a correct sequence if it is completed in less than 3 seconds. Each 3 seconds after the timer starts, the number of points awarded decreases by 30% of the points in the previous 3-second interval. Which function gives the points awarded x seconds from the start, where x is a multiple of 3?

A) $f(x) = 800(0.70)^{x/3}$
B) $f(x) = 800(0.70)^{3x}$
C) $f(x) = 800(0.30)^{x/3}$
D) $f(x) = 800(0.30)^{3x}$

Question 13

Skill: Linear Equations / Word Problem | Difficulty: Hard

A bakery produces large, medium, and small loaves. During a certain shift, the number of large loaves produced is 5 times the number n of medium loaves, and the number of small loaves is 30. During this shift, the bakery produces 120 loaves total. Which equation represents this situation?

A) $5n + 30 = 120$
B) $6n + 30 = 120$
C) $5n + n + 30 = 120$
D) $n + 30 = 120$

Question 14

Skill: Coordinate Geometry (Circles / Tangent Lines) | Difficulty: Hard

A circle in the xy-plane has its center at (7, 2). Line t is tangent to this circle at the point (a, –5), where a is a constant. The slope of line t is 4/3. What is the value of a?

Question 15

Skill: Coordinate Geometry (Distance / Area) | Difficulty: Hard

The line segment shown in the xy-plane represents one of the legs of a right triangle. The area of this triangle is $42\sqrt{17}$ square units. What is the length, in units, of the other leg of this triangle?

[PLACEHOLDER: Graph: A line segment in the xy-plane from approximately (–3, 1) to (5, 5) representing one leg of a right triangle. The area of this triangle is $42\sqrt{17}$ square units.]

A) 12
B) 14
C) $7\sqrt{17}$
D) $14\sqrt{17}$

Question 16

Skill: Geometry (Triangle Congruence) | Difficulty: Hard

In triangle MNO and triangle PQR, MN and PQ are each equal to 13 centimeters, and angles M and P each have measure 62°. Which additional piece of information is sufficient to prove that triangle MNO is congruent to triangle PQR?

A) The lengths of sides NO and QR are equal.
B) The lengths of sides MO and PR are equal.
C) The measures of angles N and Q are equal.
D) No additional information is necessary to prove that the two triangles are congruent.

Question 17

Skill: Linear Functions (Graph Interpretation) | Difficulty: Hard

The graph of the linear function $y = f(x) – 18$ is shown. If c and d are positive constants, which equation could define f?

[PLACEHOLDER: Graph: A coordinate plane showing a line with positive slope, passing through approximately (–8, –14) and (4, 16). The y-intercept of the displayed line appears to be around (0, 4).]

A) $f(x) = d – cx$
B) $f(x) = –d + cx$
C) $f(x) = d + cx$
D) $f(x) = –d – cx$

Question 18

Skill: Probability | Difficulty: Hard

For 100 insects, the table summarizes the distribution of habitat classification and body length. One of these insects will be selected at random. What is the probability of selecting an insect with a body length that is less than or equal to 30 mm, given that it is not classified as Aquatic? (Express your answer as a decimal or fraction, not as a percent.)

[PLACEHOLDER: Table with columns: Habitat, Less than 15 mm, 15 to 30 mm, Greater than 30 mm. Rows: Terrestrial: 14, 9, 6; Aquatic: 2, 11, 19; Arboreal: 10, 24, 5. Total specimens = 100.]

Question 19

Skill: Radical Equations | Difficulty: Hard

$\sqrt{(k – x)} = 36 – x$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Question 20

Skill: Factoring Polynomials | Difficulty: Hard

Which of the following expressions has a factor of $x + 5b$, where b is a positive integer constant?

A) $4x^2 + 23x + 20b$
B) $4x^2 + 24x + 20b$
C) $4x^2 + 27x + 20b$
D) $4x^2 + 29x + 20b$

Question 21

Skill: Quadratic Equations / Discriminant | Difficulty: Hard

$x(kx – 54) = –18$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

Question 22

Skill: Ratios and Percentages | Difficulty: Hard

The mass of object P is 450% of the mass of object Q, and the mass of object P is 0.060% of the mass of object R. If the mass of object R is s% of the mass of object Q, what is the value of s/1,000?

Question Count Verification

Section	Module	Count
S1 Reading & Writing	Module 1	27
S1 Reading & Writing	Module 2 — Easier	27
S1 Reading & Writing	Module 2 — Harder	27
S2 Math	Module 1	22
S2 Math	Module 2 — Easier	22
S2 Math	Module 2 — Harder	22
	TOTAL	**147**